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Vietnam's Tourism Recovery Under Climate Change and the Net Zero Transition: Mobility-Related Carbon Pressure and Short-Run Decoupling

Luong Xinh Ho[1], Thi Thu Ha Nguyen[2*]

[1] Faculty of Economics and Rural Development, Thai Nguyen University of Agriculture and Forestry, 24000 Thai Nguyen, Vietnam
[2] Faculty of Marketing and Tourism, Thai Nguyen University of Economics and Business Administration, 24000 Thai Nguyen, Vietnam

[*] Correspondence: Thi Thu Ha Nguyen (ntthuha@tueba.edu.vn)

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Abstract: Vietnam's post-pandemic tourism recovery raises the question of how renewed visitor growth relates to economic value, climate vulnerability, and mobility-related carbon performance. This study assesses Vietnam's tourism recovery under climate change and the Net Zero transition using secondary data for 2019–2023. It applies descriptive analysis, recovery indices, system-level emission-to-activity ratios, passenger-aviation Tapio decoupling analysis, and a national-transport system-level proxy elasticity for 2022–2023. National transport and passenger-aviation emissions are treated as system-level proxies for mobility-related carbon pressure and do not constitute a tourism emissions inventory. The results indicate a scale–value–carbon mismatch. In 2023, total tourist volume reached 117.3% of its 2019 level, whereas nominal tourism revenue and average nominal revenue per reported tourist visit reached 89.8% and 76.5%, respectively. Contextual climate evidence indicates vulnerability channels affecting destinations, infrastructure, transport connectivity, business operations, and service continuity. National transport and passenger-aviation emissions reached 104.1% and 93.8% of their 2019 levels. During 2022–2023, national transport emissions grew more slowly than measured passenger activity, while passenger aviation exhibited weak decoupling. Absolute emissions nevertheless increased in both systems. The results indicate short-run relative improvement in emissions performance rather than structural decarbonisation. Three policy priorities emerge: strengthening tourism carbon accounting, integrating climate adaptation into destination and transport planning, and coordinating measures to reduce mobility-related emissions. The diagnostic framework may also be relevant to tourism transitions in emerging economies facing similar climate, mobility, and data constraints.

Keywords: Climate vulnerability; Tourism recovery; Mobility-related carbon pressure; Emission-to-activity ratio; Tapio decoupling; Net Zero transition; Vietnam

1. Introduction

Climate change presents a fundamental challenge to sustainable tourism because tourism systems depend on climatic stability, ecosystem quality, infrastructure, destination accessibility, and environmental attractiveness. Rising temperatures, sea-level rise, coastal erosion, storms, flooding, seasonal change, and biodiversity loss may reduce destination competitiveness, disrupt tourism and transport services, increase adaptation costs, and affect visitor safety and experience, particularly in coastal, island, nature-based, and heritage destinations (Arabadzhyan et al., 2021; Scott et al., 2012; Scott et al., 2019). Tourism vulnerability therefore reflects not only exposure to physical hazards but also infrastructure quality, institutional capacity, business preparedness, and the adaptive capacity of workers and local communities.

Tourism also generates greenhouse gas emissions through transport, aviation, accommodation, energy use, food services, infrastructure, destination consumption, and supply chains. The global Travel and Tourism sector emitted approximately 3.41 billion tonnes of CO_2 equivalent in 2023, accounting for about 6.55% of global greenhouse gas emissions (World Travel & Tourism Council, 2024). Mobility is especially important because tourism depends on the movement of visitors between origins and destinations, creating substantial carbon pressure in systems reliant on aviation, long-distance travel, and fossil-fuel-based transport (Gössling et al., 2013; Lenzen et al., 2018; World Tourism Organization & International Transport Forum, 2019). However, national transport emissions, aviation emissions, and comprehensive tourism carbon footprints have different accounting boundaries and should not be treated as equivalent. The Net Zero transition further requires tourism vulnerability and emissions to be considered together. Net Zero entails deep reductions in gross emissions, transparent accounting, accountable governance, and credible treatment of residual emissions and removals rather than reliance on offsetting or intensity improvements alone (Fankhauser et al., 2022). Accordingly, this study defines Net Zero-aligned tourism as a transition direction based on real emissions reduction, lower-carbon mobility, value-chain governance, and improved carbon accounting, with climate resilience treated as a complementary adaptation dimension.

Vietnam provides a relevant setting for this analysis. Tourism expanded rapidly before the COVID-19 pandemic, which subsequently disrupted international travel, aviation, accommodation, and destination services. The pandemic also revealed the vulnerability of tourism systems to external shocks and highlighted the importance of assessing recovery through resilience and transformation perspectives rather than a simple return to pre-pandemic conditions (Gössling et al., 2021). Although recovery accelerated after travel restrictions were lifted, it remained uneven in 2023. Domestic tourism exceeded its pre-pandemic level, whereas international arrivals, nominal tourism revenue, and average nominal revenue per reported tourist visit remained below their 2019 levels (Vietnam National Authority of Tourism, 2024 and Vietnam National Authority of Tourism, 2025). Tourism-related employment and tourism's contribution to gross domestic product also remained below their 2019 levels (Herre & Samborska, 2023). Many major destinations are also located in coastal areas, islands, deltas, and other climate-sensitive regions exposed to sea-level rise, storms, flooding, erosion, heat stress, and ecosystem degradation (Committee on Science Technology & Environment of the National Assembly, 2017; Ministry of Natural Resources & Environment, 2025; Nguyen & Nguyen, 2022). Vietnam's commitment to achieve Net Zero emissions by 2050 further strengthens the need to move beyond volume-led recovery toward higher-value, lower-carbon, and more climate-resilient tourism development (Ministry of Industry & Trade, 2021).

Existing research has examined destination vulnerability, tourism carbon footprints, transport emissions, low-carbon tourism, and Net Zero-oriented policy (Lenzen et al., 2018; Moreno & Becken, 2009; Perch-Nielsen, 2010; World Tourism Organization & International Transport Forum, 2019). More recent studies have applied emissions-intensity, decomposition, and decoupling methods to relationships among tourism, transport activity, economic value, and emissions (Driha et al., 2025; Xu et al., 2023; Yan & Phucharoen, 2024). In Vietnam, emerging research addresses climate impacts on coastal and island tourism, tourism–emissions relationships, and the potential for Net Zero-oriented tourism development (Nguyen & Nguyen, 2022; Pham, 2025; Tran et al., 2025). Nevertheless, two gaps remain. First, tourism is commonly examined either as a climate-vulnerable system or as a source of carbon pressure, with limited integration of these dimensions in post-pandemic recovery analysis. Second, Vietnam lacks an internally consistent tourism greenhouse gas inventory separating emissions from transport, aviation, accommodation, food services, destination activities, and supply chains.

To address these gaps, the study develops an integrated diagnostic framework with two complementary dimensions. The climate-risk and resilience dimension examine climate exposure, tourism-system vulnerability, and resilience requirements affecting destinations, infrastructure, transport connectivity, businesses, workers, and local communities. The recovery–carbon dimension assesses tourism scale, market structure, nominal economic value, and mobility-related carbon performance during 2019–2023. National transport and passenger-aviation emissions are treated as system-level proxies for mobility-related carbon pressure rather than direct estimates of tourism-attributable emissions. The study addresses three questions: (1) What climate-risk exposure patterns and vulnerability channels are relevant to Vietnam's tourism destinations, infrastructure, transport connectivity, and business operations? (2) How did Vietnam's tourism scale, market structure, nominal economic value, and system-level mobility-related carbon performance change during 2019–2023, as assessed using recovery indices, emission-to-activity ratios, and short-run elasticity measures? and (3) What policy priorities emerge for a climate-resilient, lower-carbon, and Net Zero-aligned tourism transition in Vietnam?

Using secondary data for 2019–2023, the study applies descriptive analysis, recovery indices, tourism-market and economic-value indicators, system-level emission-to-activity ratios, passenger-aviation Tapio analysis, and a national-transport system-level proxy elasticity for the 2022–2023 reopening interval. Its contribution is threefold. Conceptually, it frames tourism as a climate-vulnerable system whose recovery depends on mobility systems that generate substantial carbon pressure. Empirically, it identifies a mismatch among the recovery of tourism scale, nominal economic value, and mobility-related carbon pressure. Methodologically, it combines comparisons with the 2019 baseline, system-level emission-to-activity ratios, a passenger-aviation Tapio coefficient, and a national-transport system-level proxy elasticity, while explicitly defining their accounting boundaries and limitations.

The central argument is that Vietnam recovered tourism volume faster than nominal economic value, while system-level mobility-related carbon pressure returned close to or above its pre-pandemic level. Passenger-aviation weak decoupling and slower growth in national transport emissions relative to measured passenger activity during 2022–2023 should be interpreted as short-run recovery dynamics rather than evidence of structural decarbonisation. The framework may also inform other emerging economies characterised by climate-exposed destinations, domestic-led recovery, dependence on carbon-intensive mobility, and limited tourism-specific emissions data. Section 2 develops the literature review and analytical framework, Section 3 presents the research design and methods, Section 4 reports the findings, Section 5 discusses their implications, and Section 6 concludes with policy priorities.

2. Literature Review

2.1 Climate Risks, Tourism-System Vulnerability and Adaptive Capacity

Tourism is highly sensitive to climate change because destinations, infrastructure, business operations, and visitor experiences depend on climatic stability, ecosystem quality, and resource accessibility. Rising temperatures, sea-level rise, coastal erosion, storms, flooding, biodiversity loss, and seasonal change may reduce destination attractiveness, disrupt transport and tourism services, increase operating costs, and affect visitor safety, particularly in coastal, island, nature-based, and heritage destinations (Arabadzhyan et al., 2021; Scott et al., 2012; Scott et al., 2019).

Tourism vulnerability reflects the interaction of exposure, sensitivity, and adaptive capacity. Exposure concerns contact with climate hazards, sensitivity refers to the extent to which tourism resources, infrastructure, businesses, and communities may be affected, and adaptive capacity denotes the ability to anticipate, respond to, and recover from disruption. Vulnerability therefore depends not only on physical hazards but also on infrastructure quality, institutional coordination, financial resources, business preparedness, and destination governance (Mitrică et al., 2025; Moreno & Becken, 2009; Perch-Nielsen, 2010). Strengthening tourism resilience requires coordinated action among tourism stakeholders, disaster-management systems, and local institutions to enhance adaptive capacity and response capability (Becken & Hughey, 2013).

The social dimension is equally important because tourism-dependent communities, workers, and small businesses may face income loss, business interruption, asset damage, declining resource quality, and reduced destination accessibility. Community resilience depends on local knowledge, livelihood diversity, collective action, access to resources, and participation in tourism and risk-management decisions. Local participation and institutional support can strengthen adaptive capacity during health and climate-related crises (Gabriel-Campos et al., 2021), while a just transition must consider how climate risks and transition costs are distributed among affected stakeholders (Rastegar et al., 2023).

This perspective is particularly relevant to Vietnam, where many tourism destinations are located in coastal areas, islands, deltas, and resource-dependent regions. Marine and island tourism in Northern Vietnam is exposed to sea-level rise, storms, coastal erosion, and weather variability (Nguyen & Nguyen, 2022). National assessments also identify substantial climate risks in coastal regions and cities where infrastructure and economic assets are concentrated (Committee on Science Technology & Environment of the National Assembly, 2017; Ministry of Natural Resources & Environment, 2025). Tran et al. (2025) further identified flooding, coastal water quality, mangrove coverage, tourism intensity, and disaster-response capacity as important dimensions of environmental vulnerability in tourism areas. However, existing studies remain focused mainly on individual hazards or destinations, with less attention to their implications for infrastructure, transport connectivity, businesses, workers, and local communities.

2.2 Post-Pandemic Tourism Recovery, Mobility Demand and Carbon Pressure

Post-pandemic tourism recovery is multidimensional. The return of visitor numbers does not necessarily imply equivalent recovery in revenue, value added, employment, business viability, or destination resilience. Market structure also matters because domestic and international visitors differ in expenditure, travel distance, length of stay, transport mode, and service demand. Recovery should therefore be assessed through tourism scale, market composition, economic value, and wider environmental and social outcomes rather than visitor volume alone. Global evidence similarly shows that tourism recovery has been strong but uneven and that future policy must address resilience, sustainability, inclusion, and persistent data gaps (OECD, 2024).

Mobility is central to this assessment because tourism depends on travel between origins and destinations. Tourism’s carbon footprint extends across transport, aviation, accommodation, food services, shopping, infrastructure, and supply chains. Transport-related emissions are especially important in destinations dependent on aviation, long-distance travel, private vehicles, and fossil-fuel-based mobility (Lenzen et al., 2018; Scott et al., 2008; World Tourism Organization & International Transport Forum, 2019).

Technological and intensity improvements have generally been insufficient to offset tourism-demand growth. Sun et al. (2024) find that the global tourism carbon footprint grew faster than the global economy during 2009–2019, with aviation, road transport, utilities, and private vehicles among the main sources. Their findings also show that emissions estimates vary with residence-based and destination-based accounting boundaries, reinforcing the distinction between comprehensive tourism footprints and narrower mobility indicators.

Post-pandemic trends require particular caution. Emissions reductions during COVID-19 largely reflected suppressed travel demand and service interruption rather than structural technological change. As restrictions were lifted, transport and aviation activity recovered and emissions rebounded. Aviation-dependent destinations may therefore experience renewed carbon pressure even when tourism revenue or international arrivals remain below pre-pandemic levels (Čubić & Török, 2024; Dorta Antequera et al., 2021; Jiménez-Islas et al., 2025).

Vietnam does not yet have a complete tourism carbon account covering transport, aviation, accommodation, food services, destination activities, and supply chains. National transport and passenger-aviation emissions may nevertheless provide contextual evidence on the mobility systems supporting tourism recovery, provided that they are treated as system-level proxies rather than tourism-attributable emissions.

2.3 Tourism Carbon Boundaries, Intensity Proxies and Short-Run Decoupling

The interpretation of tourism emissions depends fundamentally on the accounting boundary. A comprehensive tourism carbon footprint may include direct emissions from tourism businesses and transport, indirect emissions from purchased energy and supply chains, and emissions embodied in the goods and services consumed by visitors. Results may also differ depending on whether emissions are attributed to visitors' countries of residence, destination economies, transport operators, or flight departure locations (Lenzen et al., 2018; Sun et al., 2024). Consequently, national transport emissions, aviation emissions, and tourism carbon footprints are analytically distinct and should not be combined without harmonised boundaries.

Absolute emissions alone are insufficient for assessing relative carbon performance. Emission-intensity indicators relate emissions to an activity or economic denominator, such as passenger-kilometres, guest nights, tourist expenditure, revenue, or value added. However, the meaning of an intensity indicator depends on how closely its numerator and denominator correspond. An emissions-per-passenger-kilometre ratio is most interpretable when both variables cover the same transport modes and passenger categories. When the emissions numerator also includes freight or other non-passenger activity, the resulting ratio is a system-level proxy rather than a technical passenger-transport emission factor. Transport intensity also varies with fuel type, vehicle technology, route distance, load factor, capacity utilisation, and operating conditions (Noussan et al., 2022).

Decoupling analysis provides a complementary perspective by examining the elasticity of environmental pressure relative to an activity or economic indicator. Tapio (2005) distinguishes decoupling states according to whether emissions increase or decrease faster or more slowly than the corresponding activity. Tourism studies have applied this approach using different combinations of emissions, tourism expenditure, value added, arrivals, and transport activity (Driha et al., 2025; Xu et al., 2023; Yan & Phucharoen, 2024). However, the validity of each coefficient depends on the consistency of the paired variables and the period examined.

Short-term decoupling after a major disruption should not automatically be interpreted as structural decarbonisation. During reopening, activity and emissions may be affected by low base values, restored transport capacity, changes in route structure, passenger load factors, and temporary operating conditions. Where both variables increase and their accounting boundaries are sufficiently aligned, faster growth in passenger activity than in emissions indicates weak relative decoupling. However, this pattern does not by itself demonstrate technological progress, persistent emissions reduction, or movement toward Net Zero.

Accordingly, this study uses complementary analytical measures. First, system-level emission-to-passenger-activity ratios are used as CO_2-intensity proxies for national transport and passenger aviation during 2019–2023. Second, for the 2022–2023 reopening interval, Tapio classification is applied only to passenger aviation, while the national-transport growth-rate ratio is retained as a supplementary system-level proxy elasticity because its emissions and activity boundaries are not equivalent. Tourism scale and nominal economic value are compared separately through their recovery indices relative to 2019. This design permits an assessment of possible scale–value–carbon mismatches without treating system-level transport emissions as the carbon footprint of Vietnam's tourism sector.

2.4 Net Zero-Aligned Tourism Transition and Governance

Net Zero requires deep reductions in gross greenhouse gas emissions, with credible removals used to balance only residual emissions that are difficult to eliminate. It should not be equated with carbon offsetting or claims based solely on declining emissions intensity. Credible Net Zero strategies require clearly defined boundaries, measurable interim and long-term targets, transparent accounting, accountable governance, and explicit treatment of residual emissions and removals (Fankhauser et al., 2022).

For tourism, the transition is complex because emissions are distributed across transport, aviation, accommodation, energy, food systems, waste, infrastructure, and supply chains (Lenzen et al., 2018; World Travel & Tourism Council, 2024). A sector-wide transition therefore requires coordination among governments, destination authorities, transport operators, accommodation providers, travel businesses, energy suppliers, visitors, and local communities (Le et al., 2025; World Travel & Tourism Council, 2024). Tourism carbon accounting should distinguish direct operational emissions, purchased-energy emissions, and material value-chain emissions—corresponding broadly to Scopes 1, 2, and 3—and should be based on transparent boundaries, emissions baselines, and measurable reduction pathways rather than selected operational indicators alone (Fankhauser et al., 2022; World Travel & Tourism Council, 2024).

Aviation remains one of the most difficult components of tourism to decarbonise because available alternatives face constraints related to cost, infrastructure, energy supply, technological maturity, scalability, and implementation time. Fleet renewal, operational efficiency, sustainable aviation fuels, electrification, hydrogen, modal substitution, and demand management may all contribute, but no single measure is sufficient. Peeters & Papp (2024) show that a zero-emission tourism pathway requires systemic changes in mobility demand, transport modes, energy systems, infrastructure investment, and aviation technology. Evidence from major tourism enterprises also indicates that improvements in emissions intensity and the adoption of corporate climate targets do not necessarily reduce absolute emissions when business and travel volumes continue to grow (Gössling et al., 2024).

The transition also has important social and governance implications. Decarbonisation policies may distribute costs, benefits, employment effects, and access to tourism opportunities unevenly among stakeholder groups. Meaningful participation by local communities, workers, small tourism enterprises, and other affected groups is therefore important for transition legitimacy and for protecting tourism-dependent livelihoods. Tourism decarbonisation should strengthen local adaptive capacity rather than shift disproportionate transition costs to stakeholders with limited financial and institutional resources (Rastegar et al., 2023).

In this study, Net Zero-aligned tourism denotes a transition direction—not a verified sector-wide outcome—based on deep emissions reductions, transparent carbon accounting, lower-carbon mobility, value-chain governance, and credible treatment of residual emissions. Climate resilience is examined as a complementary adaptation dimension rather than as part of the definition of Net Zero itself. The study does not measure Vietnam's sector-wide progress toward Net Zero because the available data do not constitute a complete tourism greenhouse gas inventory.

Vietnam's commitment to achieve Net Zero emissions by 2050 provides a policy basis for aligning tourism development with the national low-carbon transition (Ministry of Industry & Trade, 2021). Existing Vietnamese studies emphasise environmental protection, resource efficiency, emissions reduction, green technologies, partnerships, and locally appropriate low-emission tourism models (Le et al., 2025; Pham, 2025). Nevertheless, empirical evidence remains limited on whether the post-pandemic recovery of tourism scale and nominal economic value has been accompanied by improved mobility-related carbon performance.

2.5 Research Gap and Analytical Framework

Existing studies commonly examine tourism either as a climate-vulnerable system or as a source of carbon emissions. This separation limits understanding of post-pandemic recovery, particularly in climate-exposed destinations that depend heavily on transport and aviation. In Vietnam, this gap is reinforced by the absence of an internally consistent tourism carbon-accounting system covering transport, accommodation, food services, destination activities, and supply chains.

To address this gap, the study develops an integrated diagnostic framework comprising two complementary dimensions. The climate-risk and resilience dimension examine climate exposure, tourism-system vulnerability, and resilience requirements affecting destinations, infrastructure, transport connectivity, businesses, workers, and local communities. Adaptive capacity and stakeholder participation are included as conceptual and policy dimensions and are considered in relation to governance, infrastructure quality, business preparedness, community participation, and institutional resources; they are not directly measured in the empirical analysis.

The recovery–carbon dimension examines tourism scale, market structure, nominal economic value, and mobility-related carbon performance during 2019–2023. National transport and passenger-aviation emissions are used as system-level proxies rather than as direct measures of tourism emissions. Recovery indices and emission-to-activity ratios are combined with passenger-aviation Tapio analysis and a national-transport system-level proxy elasticity to identify possible mismatches among tourism scale, nominal economic value, and mobility-related carbon pressure. The two dimensions are linked conceptually and contextually through climate-exposed infrastructure and transport connectivity and are considered jointly in assessing tourism-recovery quality and related transition priorities. The framework is diagnostic rather than causal: it does not estimate whether climate vulnerability caused the observed recovery or emissions patterns. Its logic may also inform other emerging economies characterised by climate-exposed destinations, domestic-led recovery, dependence on carbon-intensive mobility, and limited tourism-specific emissions data.

Figure 1 summarises the study's integrated diagnostic framework and the analytical sequence within each dimension. The framework combines a climate-risk and resilience assessment with a recovery–carbon assessment.

The first dimension identifies tourism-system vulnerability and resilience requirements, while the second examines tourism scale, nominal economic value, and mobility-related carbon performance. The two dimensions converge in an assessment of tourism-recovery quality and related climate-resilient and Net Zero-aligned transition priorities.

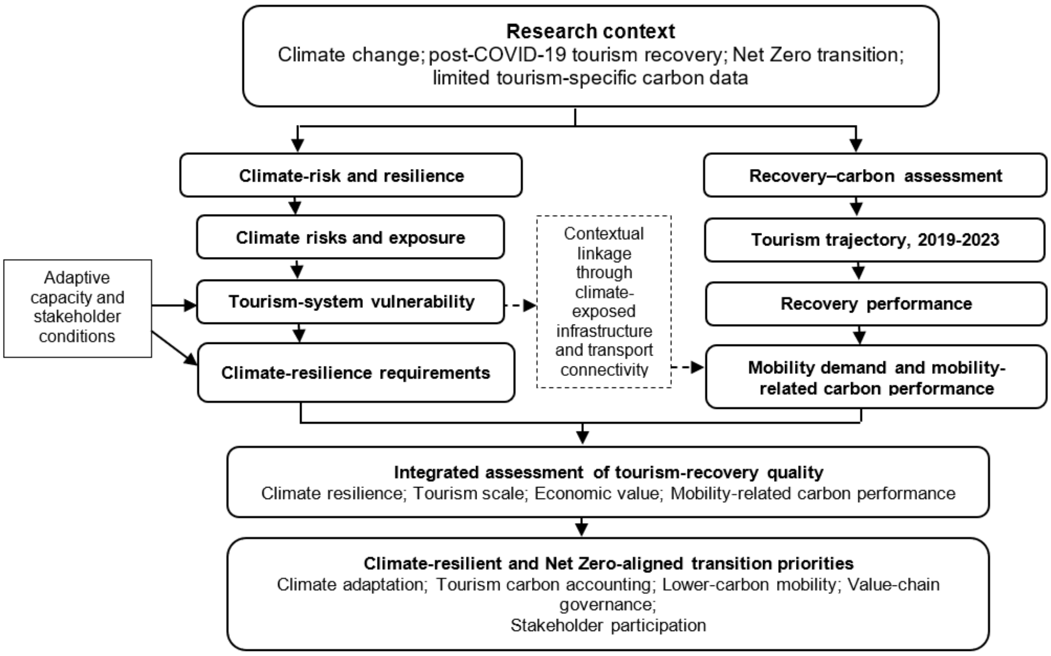

Figure 1. Integrated diagnostic framework for tourism recovery under climate change and the Net Zero transition
Note: Solid arrows show the analytical sequence within each dimension; dashed arrows denote contextual, non-causal linkages. The emissions indicators are system-level proxies. Tapio classification applies only to passenger aviation for 2022–2023, while national transport is reported as a proxy elasticity. Adaptive capacity and stakeholder participation are conceptual rather than directly measured.

3. Methodology

3.1 Research Design

This study adopts a secondary-data-based diagnostic design to examine Vietnam's tourism development under climate change and the Net Zero transition. The analytical period covers 2019–2023 and distinguishes three phases: the 2019 pre-pandemic baseline, the COVID-19 disruption during 2020–2021, and the reopening and recovery period during 2022–2023.

The research design comprises two complementary analytical dimensions. The first is the climate-risk and resilience dimension, which identifies climate-related exposure, tourism-system vulnerability, and resilience requirements affecting tourism resources, infrastructure, transport connectivity, businesses, service chains, and communities. The second is the recovery–carbon assessment dimension, which examines the recovery of tourism scale, market structure, and nominal economic value, together with changes in system-level transport and aviation emissions.

The two dimensions are linked conceptually and contextually through climate-exposed infrastructure and transport connectivity and are considered jointly in assessing tourism-recovery quality across climate resilience, visitor scale, nominal economic value, and mobility-related carbon performance. The analytical framework is diagnostic rather than causal. The study does not statistically estimate whether climate vulnerability caused the observed post-pandemic recovery pattern or changes in transport emissions.

The empirical methods include descriptive analysis, annual growth rates, recovery indices relative to the 2019 baseline, tourism-market and value indicators, system-level emission-to-activity ratios, passenger-aviation Tapio decoupling analysis, and a national-transport system-level proxy elasticity. The methods are used to identify patterns and mismatches in recovery rather than to estimate the full carbon footprint of Vietnam's tourism sector.

In this study, the scale–value–carbon mismatch refers to a diagnostic pattern in which tourism volume recovers more strongly than nominal economic value, while system-level mobility-related carbon pressure returns close to or exceeds its pre-pandemic level. It is assessed through separate indicators and does not constitute a composite index or a causal relationship.

3.2 Data Sources, Analytical Boundaries and Harmonisation

Tourism-flow, revenue, and accommodation-capacity indicators were obtained primarily from the Vietnam National Authority of Tourism (2024) and Vietnam National Authority of Tourism (2025). These include international arrivals, reported domestic tourist trips, total tourism revenue, accommodation establishments, and accommodation rooms. Total reported tourist volume was calculated by the authors as the sum of international arrivals and reported domestic tourist trips. Tourism-related employment and tourism's contribution to GDP were obtained from the Our World in Data tourism database (Herre & Samborska, 2023). Passenger-transport activity, expressed in passenger-kilometres, was obtained from the General Statistics Office of Vietnam (2026).

Climate-risk and vulnerability evidence was compiled from national climate-policy documents, regional vulnerability assessments, enterprise surveys, and tourism-related academic studies. Regional climate-vulnerability data for the Northern and Central coastal regions were extracted from a published secondary source (Committee on Science Technology & Environment of the National Assembly, 2017). The reported vulnerability classifications were used as contextual evidence and were not independently reconstructed or validated in this study. Because a consistent annual dataset measuring climate losses specifically attributable to tourism was unavailable, these sources were used to identify climate-risk exposure and vulnerability channels rather than to estimate the causal effect of climate change on annual tourism performance.

National transport CO_2 emissions were obtained from the Climate Watch dataset (Climate Watch, 2026), with major processing by Our World in Data (OWID). The transport category includes road transport, rail, domestic aviation, pipeline transport, domestic navigation, and non-specified transport. International aviation and international marine bunkers are not included in the national transport total. The indicator therefore represents the national transport system as a whole and includes passenger transport, freight, and other non-passenger activities. Aviation CO_2 emissions were obtained from the OECD Air Transport CO_2 Emissions database (OECD, 2025a; OECD, 2025b; OECD, 2025c), processed by OWID. The selected indicators cover commercial passenger flights. Domestic aviation refers to flights departing and arriving within the same country, whereas international aviation emissions are assigned to the country from which the flight departs. These data do not distinguish tourism from non-tourism passenger travel.

The transport and aviation datasets are not mutually exclusive because domestic aviation is included in the national transport-emissions series. Consequently, the two series are analysed separately and are not aggregated. Similarly, international aviation emissions assigned to departures from Vietnam are not directly matched with international tourist arrivals to Vietnam, because the two indicators represent different directions of mobility.

Estimates of the global Travel and Tourism emissions footprint published by the World Travel & Tourism Council (WTTC) are used only to provide international context. They are not combined with the national transport or aviation series because the datasets use different sectoral boundaries and accounting methods.

Across all sources, calculations were conducted within internally consistent series. Recovery indices compare observations from the same variable and source with their corresponding 2019 values. Cross-source comparisons are interpreted diagnostically and do not imply that the indicators have identical accounting boundaries.

3.3 Indicators and Analytical Methods

The scale–value–carbon mismatch was assessed through a comparative examination of three analytically distinct dimensions: tourism scale, nominal economic value, and system-level mobility-related carbon pressure. The indicators were compared separately and were not aggregated into a formal composite index. The carbon dimension represents system-level mobility-related carbon pressure rather than tourism-attributable emissions. Table 1 summarises the operational definition, data boundaries, reference periods, and permitted interpretation of the indicators used in the three dimensions.

3.3.1 Annual change and recovery indicators

Annual growth rates were calculated using Eq. (1) to describe annual changes in tourism, passenger activity, and emissions:

$$g_{X,t} = \frac{X_t - X_{t-1}}{X_{t-1}} \times 100\ (\%) \tag{1}$$

where, X_t is the value of the indicator X in year t and X_{t-1} is its value in the previous year.

Recovery relative to the pre-pandemic baseline was assessed using Eq. (2):

$$RI_{X,t} = \frac{X_t}{X_{2019}} \times 100 \tag{2}$$

where, $RI_{X,t}$ is the recovery index of indicator X in year t; and X_{2019} is its value in the pre-pandemic baseline year.

The index was calculated separately for tourism flows, nominal tourism revenue, passenger activity, and system-level emissions. For emissions indicators, the index describes the extent to which system-level carbon pressure returned to or exceeded its 2019 level and should not be interpreted as a desirable form of recovery.

Table 1. Operational definition of the scale–value–carbon mismatch

Dimension	Indicators	Data Sources	Accounting Boundary	Baseline/Analytical Period	Permitted Interpretation
Tourism scale	International arrivals; reported domestic tourist trips; total reported tourist volume; international tourist share	Vietnam National Authority of Tourism (2024) and Vietnam National Authority of Tourism (2025); authors' calculations	Reported arrivals and trips rather than counts of distinct individuals.	2019 baseline; 2019–2023 study period	Reported tourism levels, recovery, and market composition.
Nominal economic value	Nominal tourism revenue; average nominal revenue per reported tourist visit	Vietnam National Authority of Tourism (2024) and Vietnam National Authority of Tourism (2025); authors' calculations	Current-price revenue; per-visit values use reported tourist volume as the denominator.	2019 baseline; 2019–2023 study period	Nominal revenue levels and recovery; not inflation-adjusted value or individual-level expenditure.
Mobility-related carbon pressure: national transport	National transport CO_2 emissions; national transport emission-to-passenger-activity ratio; national-transport system-level proxy elasticity	Climate Watch (2026), with major processing by OWID; GSO (n.d.); authors' calculations	Emissions include passenger, freight, and other transport activities, whereas the activity denominator covers passenger-kilometres only.	2019 baseline for recovery indices; annual observations for 2019–2023; 2022–2023 for the proxy elasticity	System-level carbon pressure and exploratory relative emissions performance; not passenger-transport decoupling or tourism-attributable emissions.
Mobility-related carbon pressure: passenger aviation	Passenger-aviation CO_2 emissions; passenger-aviation emission-to-passenger-activity ratio; passenger-aviation Tapio coefficient	OECD (2025a), OECD (2025b), and OECD (2025c), processed by OWID; GSO (n.d.); authors' calculations	Commercial passenger aviation across all trip purposes; tourism and non-tourism travel are not separately identified.	2019 baseline for recovery indices; annual observations for 2019–2023; 2022–2023 for the Tapio analysis	Passenger-aviation carbon pressure and short-run decoupling; not tourism-specific decoupling or a tourism carbon footprint.

Note: The three dimensions are compared separately; no composite index is constructed. Tapio classification applies only to passenger aviation, while national transport is reported as an exploratory proxy. The two emissions series are not aggregated.

3.3.2 Tourism structure and nominal economic-value indicators

Tourism-market structure was assessed using Eq. (3):

$$IS_t = \frac{IT_t}{TT_t} \times 100 \tag{3}$$

where, IT_t represents international arrivals and TT_t represents total tourist volume, defined as the sum of international tourist arrivals and domestic tourist trips. Average nominal revenue per reported tourist visit was calculated as:

$$RPT_t = \frac{R_t}{TT_t} \tag{4}$$

where, R_t represents total nominal tourism revenue measured at current prices. The indicator therefore represents average nominal revenue per reported tourist visit and should not be interpreted as expenditure per unique individual tourist.

Because the revenue series is measured at current prices, the resulting recovery indicators represent nominal rather than real economic-value recovery. When the general price level increases, nominal recovery indices may overstate the recovery of inflation-adjusted tourism value. The reported indicators are therefore consistently referred to as nominal tourism revenue, average nominal revenue per reported tourist visit, and nominal economic-value recovery.

3.3.3 System-level transport and passenger-aviation emission-to-activity ratios

To examine how mobility-related CO_2 emissions changed relative to passenger-transport activity, the study calculated system-level emission-to-activity ratios for the national transport system and the passenger-aviation system:

$$MCI_{j,t} = \frac{E_{j,t}}{PKM_{j,t}}, j \in \{\mathrm{TR}, \mathrm{AV}\} \tag{5}$$

where, $MCI_{j,t}$ denotes the system-level emission-to-activity ratio for mobility system j in year t; $E_{j,t}$ denotes the corresponding CO_2 emissions; and $PKMj,t$ denotes passenger-transport activity measured in passenger-kilometres. The subscripts TR and AV refer to the national transport system and the passenger-aviation system, respectively.

For the national transport system, the ratio was calculated as:

$$MCI_{TR,t} = \frac{E_{TR,t}}{PKM_{TR,t}} \tag{6}$$

where, $E_{TR,t}$ represents national transport CO_2 emissions and $PKM_{TR,t}$ represents total passenger-kilometres across the national transport system.

For the passenger-aviation system, the ratio was calculated as:

$$MCI_{AV,t} = \frac{E_{AV,t}}{PKM_{AV,t}} \tag{7}$$

where, $E_{AV,t}$ represents CO_2 emissions from commercial passenger aviation and $PKM_{AV,t}$ represents air passenger-kilometres. CO_2 emissions were expressed in Mt CO_2 and passenger activity in million passenger-kilometres. Accordingly, the numerical ratio was multiplied by 10^6 to obtain g CO_2 per passenger-kilometre.

A decrease in $MCI_{j,t}$ indicates that CO_2 emissions changed less rapidly than passenger-kilometres. This may occur when passenger activity increases faster than emissions or when emissions decline faster than passenger activity. Conversely, an increase in $MCI_{j,t}$ indicates that emissions changed more rapidly than passenger activity, including cases in which emissions increase faster than passenger-kilometres or decline more slowly during a contraction. The ratios are used for temporal comparison and should not be interpreted as direct measures of tourism-attributable emissions.

In particular, $MCI_{TR,t}$ is a system-level proxy rather than a technical passenger-transport emission factor because the national transport-emissions numerator includes both passenger and non-passenger transport activities, including freight, whereas the denominator captures passenger activity only. The aviation numerator and denominator are more closely related than those of the national transport ratio, although full boundary equivalence cannot be confirmed from the available metadata. Nevertheless, the aviation indicator includes both tourism and non-tourism travel and does not distinguish passengers by trip purpose. Accordingly, neither ratio measures the full tourism carbon footprint or the carbon intensity of Vietnam's tourism sector.

Changes in these ratios are interpreted as short-run descriptive indicators of relative emissions performance. They do not, by themselves, demonstrate structural decarbonisation, technological improvement, or progress toward Net Zero. Changes in the ratios may also reflect variations in transport demand, freight activity, capacity utilisation, route composition, modal structure, operating conditions, and the exceptional base conditions associated with the post-pandemic recovery.

3.3.4 Short-run passenger-aviation Tapio decoupling and national-transport proxy elasticity

Two mathematically comparable but methodologically distinct elasticity measures were calculated for the 2022–2023 tourism-reopening period. Tapio decoupling classification was applied to passenger aviation, for which the emissions and passenger-activity variables have more closely aligned accounting boundaries. For national transport, the same growth-rate ratio was retained only as a supplementary, exploratory system-level proxy elasticity because the emissions numerator includes freight and other non-passenger activities, whereas the denominator includes passenger-kilometres only.

For passenger aviation, the Tapio decoupling coefficient was calculated as:

$$D_{AV,t} = \frac{\Delta E_{AV,t}/E_{AV,t-1}}{\Delta PKM_{AV,t}/PKM_{AV,t-1}} \quad (8)$$

where, $D_{AV,t}$ is the passenger-aviation Tapio decoupling coefficient in year t; $E_{AV,t}$ denotes passenger-aviation CO_2 emissions; and $PKM_{AV,t}$ denotes air passenger-kilometres.

For national transport, the exploratory system-level proxy elasticity was calculated as:

$$\eta_{TR,t} = \frac{\Delta E_{TR,t}/E_{TR,t-1}}{\Delta PKM_{TR,t}/PKM_{TR,t-1}} \quad (9)$$

where, $\eta_{TR,t}$ denotes the elasticity of national transport CO_2 emissions relative to total passenger activity.

Changes were calculated as:

$$\Delta E_{j,t} = E_{j,t} - E_{j,t-1} \quad (10)$$

$$\Delta PKM_{j,t} = PKM_{j,t} - PKM_{j,t-1} \quad (11)$$

Although $\eta_{TR,t}$ has the same mathematical form as a Tapio elasticity, it is not interpreted or classified as passenger-transport decoupling because its emissions and activity variables do not have consistent boundaries. It is used only as supplementary and exploratory evidence indicating whether total national transport emissions changed faster or more slowly than measured passenger activity.

The analysis was limited to the 2022–2023 interval. Earlier intervals were excluded because passenger activity and emissions were heavily affected by border closures, travel restrictions, service suspensions, and other exceptional COVID-19 conditions. The year 2019 remains the pre-pandemic baseline for the recovery-index analysis but is not treated as the starting point of a continuous Tapio series.

Following Tapio (2005), and because air passenger activity increased during 2022–2023, only the passenger-aviation coefficient was classified using the expansion-side states:

$D_{AV,t} < 0$ strong decoupling;
$0 \le D_{AV,t} < 0.8$ weak decoupling;
$0.8 \le D_{AV,t} \le 1.2$ expansive coupling;
$D_{AV,t} > 1.2$ expansive negative decoupling.

Strong decoupling occurs when air passenger activity increases while passenger-aviation emissions decline. Weak decoupling indicates that both variables increase, but emissions increase more slowly than passenger activity. Expansive coupling indicates approximately proportional growth, whereas expansive negative decoupling occurs when emissions increase faster than passenger activity. These categories were not applied to the national-transport proxy elasticity.

The passenger-aviation coefficient is interpreted only as short-run decoupling evidence for the 2022–2023 reopening interval. Because both estimates cover one exceptional annual interval, they may reflect low-base effects, restored transport capacity, and other post-pandemic operating conditions; neither provides evidence of structural decarbonisation or progress toward Net Zero. Neither measure is tourism-specific: passenger-aviation data include tourism and non-tourism travel, while national transport emissions also include freight and other non-passenger activities. Tourist arrivals and nominal tourism revenue were therefore not used as denominators. The scale–value–carbon mismatch was instead assessed comparatively using separate 2019-baseline indices and average nominal revenue per reported tourist visit.

3.4 Scope, Assumptions and Limitations

This study provides a diagnostic rather than causal assessment of Vietnam's tourism recovery, climate vulnerability, and mobility-related carbon performance during 2019–2023. As summarised in Table 1, indicators with different accounting boundaries were analysed separately and were not aggregated across incompatible series. The emissions indicators are system-level proxies and do not constitute a tourism emissions inventory.

The passenger-aviation Tapio coefficient and the national-transport system-level proxy elasticity were calculated only for the 2022–2023 reopening interval. Tapio classification was applied only to the passenger-aviation coefficient, whereas the national-transport estimate was retained as supplementary, exploratory evidence. Neither measure is tourism-specific, nor should either be interpreted as evidence of structural decarbonisation or progress toward Net Zero.

The linkage between climate vulnerability and mobility-related carbon pressure is conceptual and contextual rather than empirically tested. Community resilience and stakeholder participation are likewise included as conceptual and policy dimensions rather than directly measured variables.

4. Results

This section presents the tourism and mobility findings together with contextual evidence on climate-risk exposure and tourism-system vulnerability. The results are interpreted diagnostically rather than causally, and the mobility-emissions indicators do not represent tourism-attributable emissions.

4.1 Tourism Recovery and the Scale–Value Mismatch After COVID-19

This subsection examines Vietnam's post-pandemic tourism recovery in terms of tourist volume, market structure, nominal economic value, supply capacity, and employment. The results assess whether the recovery of visitor numbers was accompanied by a comparable recovery in tourism value and sectoral performance.

Vietnam's tourism sector experienced three distinct phases during 2019–2023: the pre-pandemic baseline in 2019, severe disruption during 2020–2021, and reopening-led recovery during 2022–2023. In 2019, Vietnam recorded 18.0 million international arrivals and 85.0 million domestic tourist trips, generating VND 755 trillion in nominal tourism revenue. COVID-19 subsequently disrupted international travel, passenger transport, accommodation, tour operations, and destination services. Recovery accelerated after travel restrictions were removed, but it remained uneven across tourism markets and economic indicators.

Table 2 indicates a strong but uneven tourism recovery. Total tourist volume reached 120.8 million in 2023, equivalent to 117.3% of the 2019 level. The recovery was primarily domestic-led: domestic tourist trips reached 127.3% of the pre-pandemic level, whereas international arrivals recovered to only 70.0%. Consequently, the international share of total tourist volume declined from 17.5% in 2019 to 10.4% in 2023. International tourism also remained highly dependent on aviation connectivity, although passenger-aviation emissions cannot be attributed exclusively to international tourists.

Table 2. Scale, market structure, supply capacity, and recovery indicators of Vietnam's tourism sector, 2019–2023

Indicator	Unit	2019	2022	2023	RI_{2023} Relative to 2019 (%)
Total tourist volume	Million reported visits	103	105	120.8	117.3
International arrivals	Million arrivals	18	3.7	12.6	70.0
Domestic tourist trips	Million trips	85	101.3	108.2	127.3
International tourist share (IS_t)	%	17.5	3.5	10.4	-
Nominal tourism revenue	VND trillion	755	495	678	89.8
Average nominal revenue per reported tourist visit (RPT_t)	VND million per reported tourist visit	7.33	4.71	5.61	76.5
Tourism contribution to GDP	%	6.8	4.3	4.8	70.6
Accommodation establishments	Thousand establishments	30	35	38	126.7
Accommodation rooms	Thousand rooms	650	700	780	120.0
Tourism-related employment	Persons/1,000 population	40	37	37	92.5

Note: Total tourist volume is the sum of international arrivals and domestic tourist trips. RPT_t is calculated as nominal tourism revenue divided by total tourist volume; revenue is reported at current prices. Tourism's contribution to GDP is reported as a percentage of GDP, while tourism-related employment is reported as the number of persons employed in tourism-related industries per 1,000 population, following Herre & Samborska (2023). A dash (-) denotes no data.

Source: Vietnam National Authority of Tourism (2024) and Vietnam National Authority of Tourism (2025) for tourism flows, nominal tourism revenue, and accommodation indicators; Herre & Samborska (2023) for tourism's contribution to GDP and tourism-related employment; authors' calculations for total tourist volume, IS_t, RPT_t, and $RI_{X,2023}$.

Recovery in visitor volume outpaced recovery in economic value. Nominal tourism revenue, average nominal revenue per reported tourist visit, and tourism's contribution to GDP reached only 89.8%, 76.5%, and 70.6% of their 2019 levels, respectively. Because the revenue indicators are measured at current prices, they should not be interpreted as measures of inflation-adjusted tourism-value recovery. Meanwhile, accommodation establishments and rooms increased to 126.7% and 120.0% of their pre-pandemic levels, while tourism-related employment remained below the 2019 baseline.

Overall, Vietnam's recovery was domestic-led and volume-oriented but economically incomplete, revealing a scale–value mismatch between visitor growth and the recovery of nominal economic value.

4.2 Contextual Evidence on Climate-Risk Exposure and Tourism-System Vulnerability

This subsection presents contextual evidence on climate-risk exposure and vulnerability channels relevant to Vietnam's tourism system, drawing on spatial vulnerability estimates and enterprise-reported disruptions.

Many Vietnamese tourism destinations, transport connections, settlements, and tourism-dependent communities are located in coastal areas, islands, deltas, and nature-dependent regions. These areas may be exposed to storms, sea-level rise, flooding, coastal erosion, prolonged heat, and ecosystem degradation, with potential implications for destination accessibility, infrastructure, business continuity, workers, and local communities. The spatial distribution of climate vulnerability across Vietnam's coastal regions provides important contextual evidence for understanding the potential exposure of tourism destinations and related infrastructure to climate risks (Table 3).

Table 3. Spatial distribution of climate vulnerability in the Northern and Central coastal regions of Vietnam

Coastal Region	Low Vulnerability (%)	Medium Vulnerability (%)	Relatively High Vulnerability (%)	High Vulnerability (%)	Relatively High + High (%)
Northern coastal region	7.60	40.86	35.12	16.42	51.54
Central coastal region	25.82	42.28	17.84	14.06	31.90

Note: Percentages represent the shares of the assessed study area in each vulnerability category, based on the index-value ranges reported in the source study. The combined "relatively high + high" category was calculated by the authors. The indicators do not measure tourism-specific damage.
Source: Committee on Science Technology & Environment of the National Assembly (2017); authors' calculations for the "relatively high + high" category.

According to the source assessment, 51.54% of the assessed area in the Northern coastal region was classified as having relatively high or high vulnerability, compared with 31.90% in the Central coastal region. The Central coastal region also had a substantial medium-vulnerability share of 42.28%, indicating that vulnerability extended across a considerable proportion of the assessed area. These spatial patterns are relevant to tourism because the coastal regions contain tourism destinations, heritage sites, transport corridors, urban centres, and service infrastructure. However, the indicators do not directly quantify the exposure or losses of these tourism-related assets; rather, they provide contextual evidence for tourism adaptation and resilience planning.

Climate risks may also affect tourism through operational and business channels. Table 4 presents reported climate phenomena and business impacts from a survey of 10,356 enterprises in 63 provinces and cities conducted by the Vietnam Chamber of Commerce & Industry (2020). Although the survey did not identify tourism enterprises separately, the findings provide contextual evidence relevant to tourism because tourism operations depend on transport access, labour productivity, physical infrastructure, logistics, energy, water, and service continuity.

Table 4. Reported climate phenomena and business-impact channels relevant to tourism-system vulnerability

Indicator Group	Indicator	Reported Share/Value
Observed climate phenomena	More prolonged heat	92.3%
	Heavy rain with storms or tropical depressions	79.9%
	Flooding in areas where it rarely occurred before	70.6%
	More frequent drought	62.0%
	Tidal flooding	53.8%
	Coastal erosion	47.9%
Business impacts	Business interruption	54.0%
	Reduced labour productivity	51.0%
	Revenue decline	51.0%
	Disrupted transport channels	46.0%
	Increased production and business costs	44.0%
	Reduced product or service quality	37.0%
	Damage to physical assets	34.0%
Disruption and loss	Median number of interrupted days	7 days/year
	Median financial loss	VND 20 million/year

Note: Percentages indicate the share of surveyed enterprises reporting each phenomenon or impact. The survey covers enterprises across sectors and is not a tourism-industry survey. Interruption days and financial losses are median reported values.
Source: Vietnam Chamber of Commerce & Industry (2020); authors' interpretation.

Among the reported climate phenomena, prolonged heat was the most prevalent, cited by 92.3% of surveyed enterprises. Heavy rain associated with storms or tropical depressions was reported by 79.9%, while 70.6% reported flooding in areas where it had rarely occurred previously. More frequent drought, tidal flooding, and coastal erosion were also widely reported.

The operational impacts were similarly broad. Business interruption was reported by 54.0% of enterprises, reduced labour productivity and revenue decline by 51.0%, disrupted transport channels by 46.0%, and increased production and business costs by 44.0%. Other reported impacts included reduced product or service quality and damage to physical assets. The median reported interruption was seven days per year, and the median financial loss was VND 20 million per year.

For tourism, these channels are relevant to passenger access, accommodation operations, tour scheduling, facility maintenance, service quality, labour conditions, and destination experience. Transport disruption may also weaken the connectivity on which tourism recovery depends, thereby providing a contextual, rather than empirically tested, link between climate vulnerability and mobility systems. Nevertheless, Tables 3 and 4 do not measure tourism-specific climate damages, community resilience, stakeholder participation, or the causal effects of climate risks on visitor numbers and emissions. Their role in the analysis is to provide contextual evidence on exposure patterns and vulnerability channels that should be considered when evaluating the quality and resilience of tourism recovery.

4.3 System-Level Mobility-Related CO_2 Emissions During 2019–2023

National transport and passenger-aviation CO_2 emissions are examined separately as system-level indicators of mobility-related carbon pressure during 2019–2023. Because the two series have overlapping accounting boundaries, they are not aggregated. International aviation emissions are also not directly matched with inbound international arrivals because the indicators follow different mobility directions and accounting conventions.

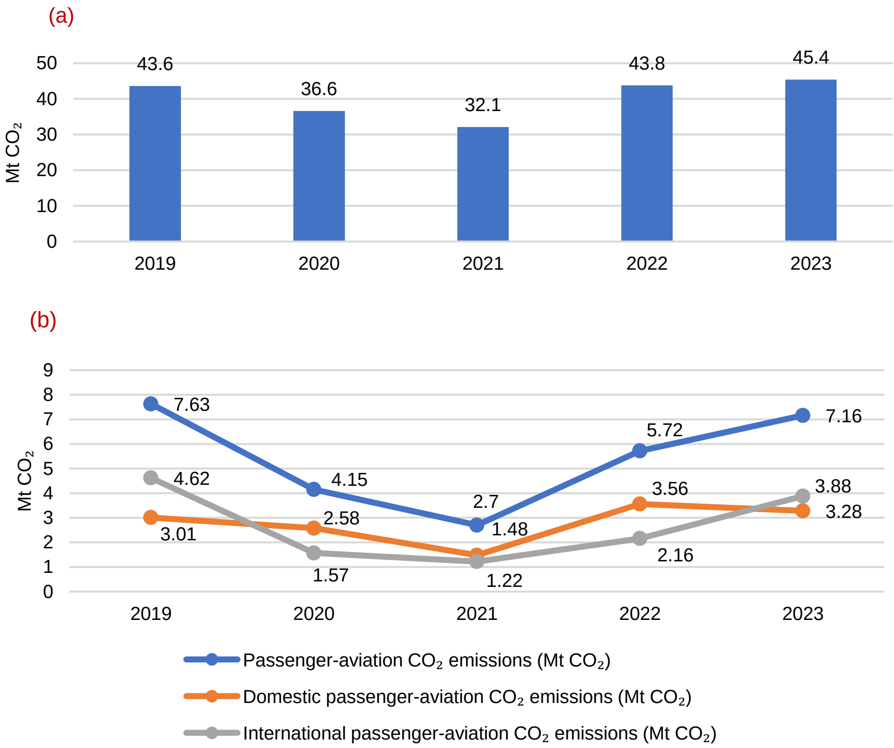

Figure 2. National transport and passenger-aviation CO_2 emissions in Vietnam, 2019–2023: (a) national transport CO_2 emissions; (b) passenger-aviation CO_2 emissions

Note: Panel (a) presents national transport CO_2 emissions. Panel (b) presents total, domestic, and international passenger-aviation CO_2 emissions. Each panel uses a single vertical axis measured in Mt CO_2. The national transport and passenger-aviation series have overlapping accounting boundaries and are not additive.

Source: Climate Watch (2026), OECD (2025a), OECD (2025b), and OECD (2025c), with data processing by Our World in Data.

Figure 2 shows that national transport emissions declined from 43.60 Mt CO_2 in 2019 to 36.60 Mt CO_2 in 2020 and 32.10 Mt CO_2 in 2021. They subsequently rebounded to 43.80 Mt CO_2 in 2022 and 45.40 Mt CO_2 in 2023, exceeding the pre-pandemic baseline.

Passenger-aviation emissions contracted more sharply, falling from 7.63 Mt CO_2 in 2019 to 4.15 Mt CO_2 in 2020 and 2.70 Mt CO_2 in 2021. They then increased to 5.72 Mt CO_2 in 2022 and 7.16 Mt CO_2 in 2023. The large percentage rebound in 2022 was partly amplified by the low 2021 base.

Domestic and international passenger-aviation emissions followed different recovery patterns. Domestic emissions declined from 3.01 Mt CO_2 in 2019 to 1.48 Mt CO_2 in 2021, rebounded to 3.56 Mt CO_2 in 2022, and decreased slightly to 3.28 Mt CO_2 in 2023. International emissions fell from 4.62 Mt CO_2 in 2019 to 1.22 Mt CO_2 in 2021 before recovering to 3.88 Mt CO_2 in 2023, remaining below their pre-pandemic level. These patterns should not be mapped directly onto domestic-tourist or international-arrival statistics because the tourism and aviation series have different populations and accounting boundaries.

Overall, the pandemic-disruption period coincided with a temporary decline in mobility-related emissions, followed by a rapid rebound during reopening. The return of emissions toward or above their 2019 levels is consistent with temporary activity suppression rather than structural decarbonisation.

4.4 Recovery Comparison and Mobility-Related Emissions Performance

4.4.1 Recovery indices relative to the 2019 baseline

Recovery indices compare selected tourism, nominal economic-value, and system-level mobility-related emissions indicators with their respective 2019 baselines (Table 5).

Table 5. Recovery indices of system-level mobility-related CO_2 emissions in Vietnam, 2023 relative to 2019

Indicator	CO_2 Emissions in 2019 (Mt CO_2)	CO_2 Emissions in 2023 (Mt CO_2)	RI_{2023} Relative to 2019 (%)
National transport CO_2 emissions	43.60	45.40	104.13
Passenger-aviation CO_2 emissions	7.63	7.16	93.84
Domestic passenger-aviation CO_2 emissions	3.01	3.28	108.97
International passenger-aviation CO_2 emissions	4.62	3.88	83.98

Note: The national transport and passenger-aviation series have overlapping accounting boundaries and are not additive.
Source: Climate Watch (2026), OECD (2025a), OECD (2025b), and OECD (2025c), processed by Our World in Data.

By 2023, total tourist volume had reached 117.3% of its 2019 level, whereas nominal tourism revenue and average nominal revenue per reported tourist visit had recovered to only 89.8% and 76.5%, respectively. Over the same period, national transport emissions reached 104.13% of their 2019 level, while passenger-aviation emissions reached 93.84%. Taken together, these separate indicators indicate a scale–value–carbon mismatch: tourism volume exceeded the pre-pandemic baseline, but nominal economic value did not recover proportionally. Meanwhile, national transport emissions surpassed their 2019 level, and passenger-aviation emissions nearly returned to it. Domestic passenger-aviation emissions reached 108.97% of their 2019 level, whereas international passenger-aviation emissions reached 83.98%. These segment-specific indices describe 2023 aviation-emissions levels relative to 2019 and should not be directly equated with the recovery of domestic or international tourist flows.

Overall, tourism volume recovered more strongly than nominal economic value, while system-level mobility-related emissions returned close to or above their pre-pandemic levels.

4.4.2 System-level emission-to-activity ratios

System-level emission-to-activity ratios compare CO_2 emissions with measured passenger activity in passenger-kilometres. Within each mobility system, lower values indicate lower emissions relative to measured passenger activity, whereas higher values indicate the opposite. For national transport, however, the ratio is interpreted only as a system-level proxy because the emissions and activity variables have different accounting boundaries (Table 6).

Table 6. System-level national-transport and passenger-aviation emission-to-activity ratios in Vietnam, 2019–2023

Mobility System	Indicator	Unit	2019	2020	2021	2022	2023
National transport	CO_2 emissions, $E_{TR,t}$	Mt CO_2	43.60	36.60	32.10	43.80	45.40
	Passenger activity, $PKM_{TR,t}$	Million passenger-km	230,747.19	151,100.54	93,805.28	170,443.72	227,328.11
	System-level proxy emission-to-activity ratio, $MCI_{TR,t}$	g CO_2/passenger-km	189.0	242.2	342.2	257.0	199.7
Passenger aviation	CO_2 emissions, $E_{AV,t}$	Mt CO_2	7.63	4.15	2.70	5.72	7.16
	Air passenger activity, $PKM_{AV,t}$	Million passenger-km	77,402.83	34,124.88	13,932.58	57,716.10	81,144.57
	Emission-to-activity ratio, $MCI_{AV,t}$	g CO_2/passenger-km	98.6	121.6	193.8	99.1	88.2

Note: $MCI_{TR,t}$ is a system-level proxy because it pairs total national transport CO_2 emissions with passenger-kilometres; $MCI_{AV,t}$ covers passenger aviation across all trip purposes.
Source: Passenger activity data from GSO (n.d.); emissions data from Climate Watch (2026), OECD (2025a), OECD (2025b), and OECD (2025c), processed by OWID.

The national transport proxy ratio increased from 189.0 g CO_2 per passenger-kilometre in 2019 to 342.2 in 2021 because measured passenger activity contracted more sharply than total national transport emissions. It subsequently declined to 199.7 in 2023 but remained approximately 5.7% above its 2019 level. Because of the boundary mismatch, this change should not be interpreted as a technical measure of passenger-transport efficiency.

The passenger-aviation ratio increased from 98.6 g CO_2 per passenger-kilometre in 2019 to 193.8 in 2021, before declining to 88.2 in 2023. The 2023 value was approximately 10.5% below its 2019 level, indicating that, relative to 2019, air passenger-kilometres recovered more strongly than passenger-aviation emissions.

The decline in the passenger-aviation ratio represents an improvement in emissions relative to measured passenger activity, but it does not by itself demonstrate structural aviation decarbonisation.

4.4.3 Short-run passenger-aviation decoupling and national-transport proxy elasticity during the recovery period

During the 2022–2023 reopening interval, passenger activity increased faster than CO_2 emissions in both mobility systems. However, only the passenger-aviation result was classified under the Tapio framework, whereas the national-transport estimate was treated as a system-level proxy elasticity (Table 7).

Table 7. Short-run passenger-aviation Tapio result and national-transport proxy elasticity, 2022–2023

Mobility System and Activity–Emissions Pair	Passenger-Activity Growth (%)	CO_2 Emissions Growth (%)	Elasticity Measure	Interpretation
National transport: total passenger-kilometres and national transport CO_2 emissions	33.37	3.65	$\eta_{TR} = 0.11$	Emissions increased more slowly than measured passenger activity
Passenger aviation: air passenger-kilometres and passenger-aviation CO_2 emissions	40.59	25.17	$D_{AV} = 0.62$	Weak decoupling

Note: Growth rates refer to changes from 2022 to 2023. Tapio classification applies only to passenger aviation; the national-transport measure is a system-level proxy elasticity. Neither measure is tourism-specific.

Source: Passenger activity data from GSO (n.d.); emissions data from Climate Watch (2026), OECD (2025a), OECD (2025b), and OECD (2025c), processed by OWID.

For national transport, passenger-kilometres increased by 33.37%, while total transport CO_2 emissions increased by 3.65%, yielding a system-level proxy elasticity of 0.11. This indicates that total national transport emissions increased more slowly than measured passenger activity. The estimate is not classified as passenger-transport decoupling. In passenger aviation, passenger-kilometres increased by 40.59%, while emissions increased by 25.17%, resulting in a Tapio coefficient of 0.62 and a classification of weak decoupling. Because both passenger activity and emissions increased, this represents relative rather than absolute decoupling.

Both measures cover only the 2022–2023 reopening interval and may reflect low-base and post-pandemic recovery effects. They should therefore be interpreted as short-run evidence rather than as indicators of structural decarbonisation or progress toward Net Zero.

4.5 Integrated Findings on Climate Vulnerability and Recovery–Carbon Performance

The climate-risk findings provide contextual information on spatial vulnerability patterns and operational vulnerability channels relevant to Vietnam's tourism system. The assessed coastal areas relevant to tourism contain substantial proportions classified at medium to high vulnerability levels, while the enterprise evidence points to business interruption, transport disruption, reduced labour productivity and revenue, higher operating costs, declining service quality, and damage to physical assets. These findings indicate potential vulnerability channels affecting destinations, infrastructure, transport connectivity, and business operations. However, they neither quantify tourism-specific climate losses nor establish causal effects on annual tourism recovery.

The recovery–carbon findings reveal an uneven relationship among tourism scale, market structure, nominal economic value, and system-level mobility-related carbon performance. Total tourist volume exceeded its 2019 level, driven primarily by domestic tourism, whereas international arrivals and nominal economic-value indicators remained below their pre-pandemic baselines. National transport CO_2 emissions exceeded the 2019 level, while passenger-aviation emissions approached it. During 2022–2023, passenger activity increased faster than emissions in both mobility systems. For national transport, this produced a system-level proxy elasticity indicating that total transport emissions increased more slowly than measured passenger activity. For passenger aviation, the Tapio coefficient indicated weak decoupling. Absolute emissions nevertheless increased in both systems.

Taken together, the findings indicate a scale–value–carbon mismatch: tourism scale recovered more strongly than nominal economic value, while mobility-related carbon pressure remained substantial. The passenger-aviation weak decoupling observed during 2022–2023 indicates a short-run relative improvement in emissions performance rather than structural decarbonisation. The national-transport result provides only supplementary system-level evidence and is not interpreted as passenger-transport decoupling.

5. Discussion

This study indicates that Vietnam's post-pandemic tourism recovery occurred under two concurrent pressures: continued exposure to climate risks and dependence on mobility systems that generate substantial carbon emissions. The findings therefore support a broader assessment of tourism recovery that considers not only visitor volume and economic value, but also climate vulnerability and system-level mobility-related carbon performance.

5.1 Domestic-Led Recovery and the Scale–Value Mismatch

Vietnam's tourism recovery was primarily driven by domestic travel, while international arrivals and nominal economic-value indicators recovered more slowly. Domestic tourism was less dependent on the restoration of international air routes, cross-border regulations, visa procedures, and conditions in origin markets. Pent-up demand and the temporary substitution of outbound travel with domestic trips may also have contributed to the rapid rebound. These mechanisms are plausible explanations, although they cannot be tested directly with the aggregate data used in this study.

The recovery of visitor volume was not accompanied by a proportional recovery in average nominal revenue per reported tourist visit or tourism's contribution to GDP. This indicates a scale–value mismatch in which reported tourist volume recovered more strongly than the economic value associated with that recovery. Such uneven recovery patterns are consistent with broader evidence that COVID-19 generated structural disruptions in tourism systems and highlighted the importance of assessing recovery through resilience and transformation perspectives rather than a simple return to pre-pandemic conditions (Gössling et al., 2021). The shift toward domestic tourism may partly explain this pattern because domestic and international visitors differ in length of stay, accommodation use, travel distance, and expenditure behaviour. Because the revenue indicators are measured at current prices, they should not be interpreted as measures of inflation-adjusted tourism-value recovery.

This finding is consistent with studies showing that visitor-volume-led tourism does not necessarily produce corresponding improvements in economic value or environmental performance (Gössling et al., 2013; Lenzen et al., 2018). The Vietnam case extends this argument by showing that post-crisis recovery may be quantitatively strong but remain incomplete in terms of value creation. The continued expansion of accommodation capacity, despite weaker recovery in value and employment indicators, further suggests that supply growth and tourist numbers alone are insufficient measures of recovery performance.

5.2 Climate Vulnerability, Mobility Dependence, and Carbon Pressure

The contextual climate evidence indicates that vulnerability is not limited to the degradation of beaches, ecosystems, or other tourism resources. Storms, flooding, coastal erosion, sea-level rise, and prolonged heat may also affect infrastructure, destination accessibility, labour productivity, business continuity, operating costs, and service quality. This is consistent with previous research identifying climate, ecosystems, landscapes, water resources, and perceived safety as important determinants of coastal destination attractiveness (Moreno & Becken, 2009; Perch-Nielsen, 2010; Scott et al., 2019).

The findings are also consistent with the climate-impact-chain perspective of Arabadzhyan et al. (2021), in which hazards affect tourism through interconnected exposure and vulnerability channels. In Vietnam, transport connectivity provides a relevant conceptual link between climate vulnerability and tourism recovery. Roads, airports, ports, and local transport systems enable visitor access and tourism supply chains, but they are themselves exposed to climate-related disruption. Climate-related disruption may reduce the reliability of these networks and require rerouting, service restoration, or additional resilience investment. These responses may affect operating conditions and mobility-related carbon pressure, although this relationship is not tested empirically in the present study. Mobility infrastructure also connects climate resilience with carbon pressure. Transport and aviation emissions were not treated in this study as a complete inventory of tourism emissions, but as system-level proxies for the mobility systems on which tourism depends. This distinction is important because transport, particularly aviation, constitutes a major component of tourism-related emissions globally (Scott et al., 2008; World Tourism Organization & International Transport Forum, 2019). Dorta Antequera et al. (2021) similarly show that aviation-dependent destinations face a structural tension between connectivity-driven tourism growth and emissions reduction.

Climate resilience and carbon mitigation should therefore not be treated as separate objectives. Infrastructure designed only to restore or expand connectivity may reinforce carbon-intensive mobility, while low-carbon infrastructure may remain vulnerable if future climate conditions are not incorporated into planning. The present study identifies this relationship conceptually and contextually; it does not establish a causal effect of climate hazards on annual mobility emissions or tourism recovery.

5.3 Short-Run Decoupling and the Net Zero Transition

The rebound in transport and passenger-aviation emissions is consistent with the reductions observed during 2020–2021 being driven mainly by restricted mobility rather than structural decarbonisation. As passenger activity recovered, absolute emissions increased in both systems. The emission-to-activity ratios require careful interpretation. For national transport, the emissions numerator includes freight and other non-passenger activities, whereas the denominator contains passenger-kilometres; the ratio is therefore a system-level diagnostic rather than a measure of passenger-transport efficiency. In aviation, passenger-kilometres recovered faster than emissions, potentially reflecting changes in load factors, route structure, aircraft deployment, and capacity utilisation. This interpretation is consistent with Noussan et al. (2022), who show that transport emission intensity depends on technology, fuels, operational conditions, and accounting boundaries.

For passenger aviation, the Tapio coefficient indicates weak relative decoupling during 2022–2023 because air passenger activity grew faster than passenger-aviation emissions. For national transport, the system-level proxy elasticity similarly indicates that total transport emissions grew more slowly than measured passenger activity, but this result is not classified as passenger-transport decoupling. Emissions nevertheless increased in both systems, and the analysis covers an exceptional reopening period affected by low-base effects and network restoration. Both measures therefore represent short-run recovery dynamics rather than structural decarbonisation. This cautious interpretation is consistent with previous studies showing that decoupling outcomes vary across time periods, activity indicators, transport structures, and emissions boundaries (Driha et al., 2025; Yan & Phucharoen, 2024; Yang & Jia, 2022).

From a Net Zero perspective, relative improvement is insufficient without sustained absolute emissions reductions and changes in fuels, technologies, transport modes, infrastructure, and travel demand. This accords with Fankhauser et al. (2022), who emphasise real and verifiable emissions reductions, and Peeters & Papp (2024), who identify systemic transformation in mobility and energy systems as essential to tourism decarbonisation. Although specific to Vietnam, the findings may be relevant to other emerging economies where climate-exposed tourism depends heavily on carbon-intensive mobility. They also imply that workers, small businesses, and local communities should be considered in designing a fair transition, although these social effects were not directly measured in this study.

Overall, the evidence indicates weak relative decoupling in passenger aviation and slower emissions growth than measured passenger activity at the national-transport system level, but not a stable Net Zero-aligned transition. This distinction highlights the need for stronger tourism carbon accounting, lower-carbon mobility, and climate-resilient infrastructure planning.

6. Conclusions and Policy Implications

This study examined Vietnam's post-pandemic tourism recovery through the complementary dimensions of climate vulnerability, economic recovery, and system-level mobility-related carbon pressure. The contextual evidence indicates that climate risks may affect not only coastal resources and destinations but also infrastructure, transport connectivity, business operations, and service continuity. Tourism vulnerability should therefore be considered a systemic development risk. Vietnam's recovery was strong in scale but uneven in composition and value. Total tourist volume exceeded the 2019 level, mainly because of domestic tourism, whereas international arrivals and nominal economic-value indicators remained below their pre-pandemic baselines. Meanwhile, national transport emissions exceeded the 2019 level and passenger-aviation emissions approached it. During 2022–2023, total national transport emissions grew more slowly than measured passenger activity, while passenger aviation exhibited weak decoupling. Absolute emissions nevertheless increased in both systems. The passenger-aviation weak decoupling therefore reflects short-run reopening dynamics rather than structural decarbonisation.

The findings identify a scale–value–carbon mismatch: tourism scale recovered more strongly than nominal economic value, while mobility-related carbon pressure remained substantial. Three policy priorities follow. First, Vietnam should establish a tourism carbon-accounting system that distinguishes tourism-attributable emissions across transport, aviation, accommodation, food services, destination activities, and supply chains. Second, climate adaptation and emissions reduction should be integrated into coastal destination and transport-infrastructure planning. Third, government, tourism businesses, transport providers, and local stakeholders should support higher-value tourism, lower-carbon mobility, operational efficiency, transparent emissions reporting, and inclusive transition measures. Government agencies could adapt the framework as an annual monitoring dashboard combining tourism scale, economic value, passenger activity, mobility emissions, and climate-risk indicators, subject to the availability of consistent annual and tourism-specific data. Tourism businesses and transport providers could use it to establish emissions baselines, assess infrastructure investments against climate risks, improve fleet and route efficiency, and report progress against measurable reduction targets.

The study contributes an integrated diagnostic framework that brings together tourism recovery, climate vulnerability, and mobility-related carbon performance. It also demonstrates how recovery indices, emission-to-activity ratios, passenger-aviation Tapio analysis, and a national-transport system-level proxy elasticity can provide an initial assessment where tourism-specific carbon accounts are unavailable. The numerical recovery and emissions results are specific to Vietnam, whereas the diagnostic logic may apply to other emerging economies where climate-exposed destinations, domestic-led recovery, carbon-intensive mobility, and limited tourism-specific emissions data coexist.

The conclusions remain subject to several limitations. National transport and passenger-aviation emissions are used as system-level indicators of mobility-related carbon pressure rather than as a complete tourism emissions inventory. National transport emissions include freight and other non-passenger activities, while passenger-aviation emissions include both tourism and non-tourism travel. The passenger-aviation Tapio coefficient and the national-transport system-level proxy elasticity cover only the exceptional 2022–2023 reopening interval. Climate evidence is contextual rather than causal, and tourism revenue is measured in nominal terms. Future research should develop tourism-attributable carbon accounts, use inflation-adjusted indicators and longer time series, and incorporate destination-, firm-, worker-, and community-level data to assess progress toward a durable climate-resilient and Net Zero-aligned tourism pathway.

Author Contributions

Conceptualization, T.T.H.N.; methodology, T.T.H.N.; software, L.X.H.; formal analysis, L.X.H.; data curation, L.X.H. and T.T.H.N.; writing—original draft preparation, L.X.H. and T.T.H.N.; writing—review and editing, L.X.H. and T.T.H.N.; visualization, L.X.H.; supervision, T.T.H.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The source data are publicly available from the sources cited in Section 3.2. The processed dataset and calculation files are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used generative AI only for language editing and improving readability. All conceptualization, analysis, and conclusions were conducted by the authors, who take full responsibility for the content of the manuscript.

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